

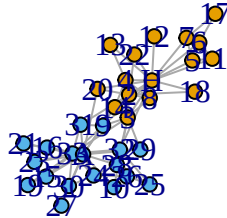
# Soc 111 - Week 5 Tuesday - Community Detection

GR

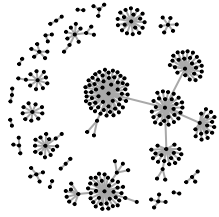
2024-09-02

```
library(igraph)
library(igraphdata)
library(tidyverse)
```

```
data("karate")
karate.g <- upgrade_graph(karate) # load Zachary's karate club
karate.layout <- layout_with_fr(karate.g)
plot(karate.g, layout=karate.layout)
```



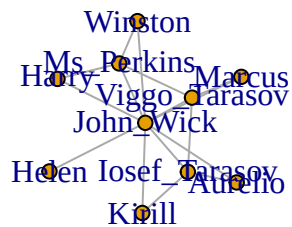
```
iliad.g <- read_graph("iliad_killed.net", format = "pajek") %>% # load the Iliad combat network
  as.undirected() %>% # force to be undirected
  igraph::simplify() # eliminate redundant edges
iliad.layout <- layout_with_fr(iliad.g)
plot(iliad.g,
     layout=iliad.layout,
     vertex.label=NA,
     vertex.size=2)
```



```
jw.df <- read_csv("johnwick.csv", col_names = F)

## Rows: 17 Columns: 2
## -- Column specification -----
## Delimiter: ","
## chr (2): X1, X2
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

jw.g <- jw.df %>% graph_from_data_frame(directed = F) %>% igraph::simplify()
jw.layout <- layout_with_fr(jw.g)
plot(jw.g, layout=jw.layout)
```



Maximizing, granularity, “search”

## Modularity

This section lightly edited from <https://r.igraph.org/reference/modularity.igraph.html>

$$Q = \frac{1}{2m} \sum_{i,j} (A_{ij} - \gamma \frac{k_i k_j}{2m}) \delta(c_i, c_j),$$

Where

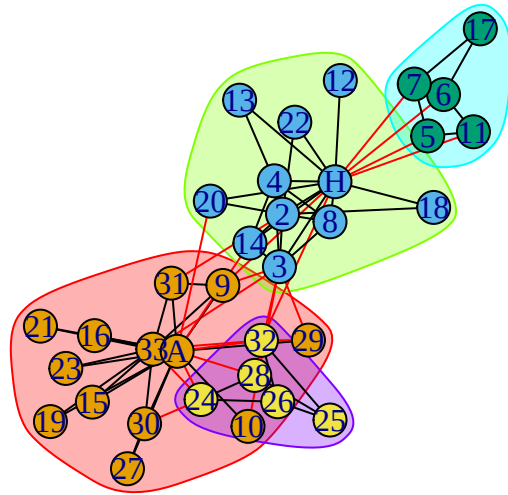
- $m$  is the number of edges
- $A_{ij}$  is a cell in the  $A$  adjacency matrix defining an edge between node  $i$  and node  $j$
- $k_i$  is degree of node  $i$
- $k_j$  is the degree of node  $j$
- $c_i$  is the type (or component) of  $i$
- $c_j$  that of  $j$
- $\Sigma$  or sum goes over all combinations of vertices
- $\delta(x, y)$  is 1 if  $x = y$  and 0 otherwise

## Walktrap

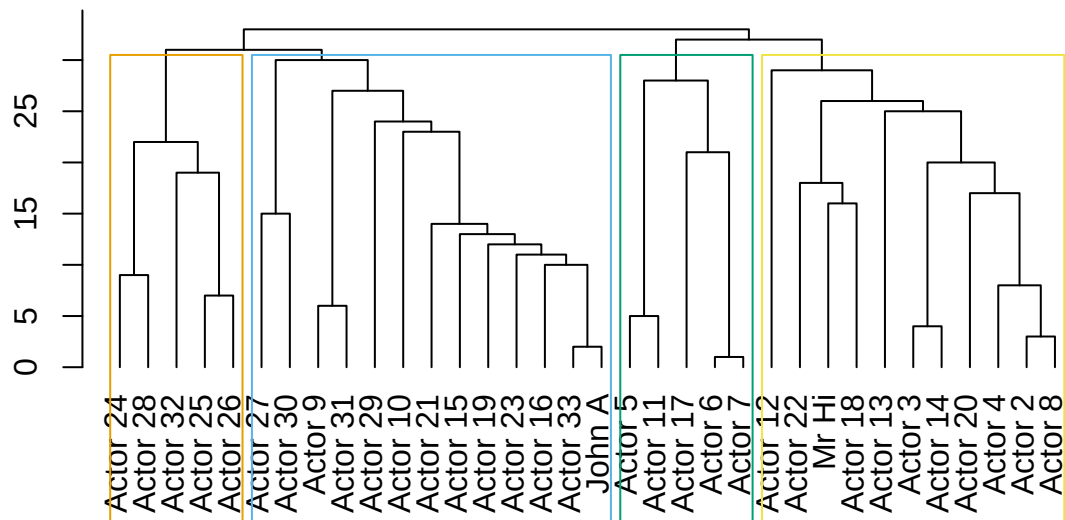
```
karate.comm.wt <- karate.g %>% cluster_walktrap()  
modularity(karate.comm.wt)
```

```
## [1] 0.440181
```

```
plot(karate.comm.wt, karate.g, layout=karate.layout)
```



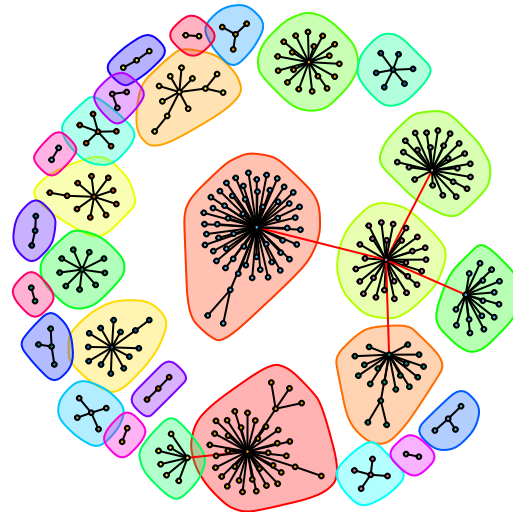
```
karate.comm.wt %>% plot_dendrogram()
```



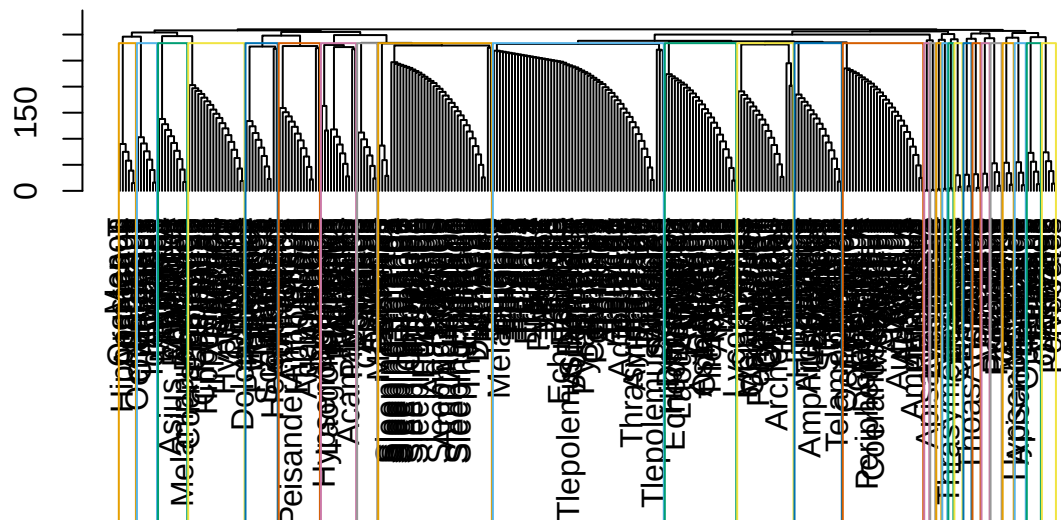
```
iliad.comm.wt <- iliad.g %>% cluster_walktrap()
modularity(iliad.comm.wt)
```

```
## [1] 0.8922707
```

```
plot(iliad.comm.wt,
      iliad.g,
      layout=iliad.layout,
      vertex.label=NA,
      vertex.size=2)
```



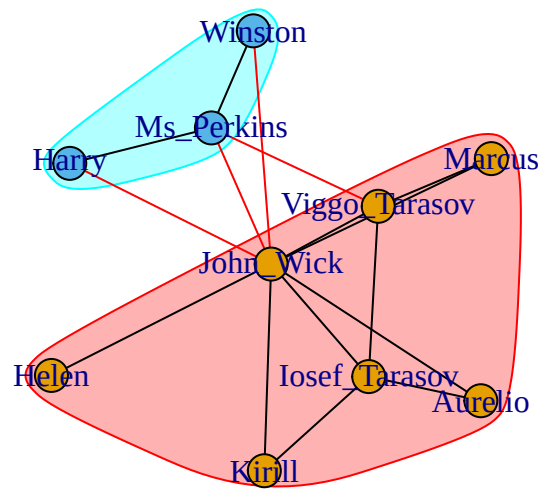
```
iliad.comm.wt %>% plot_dendrogram()
```



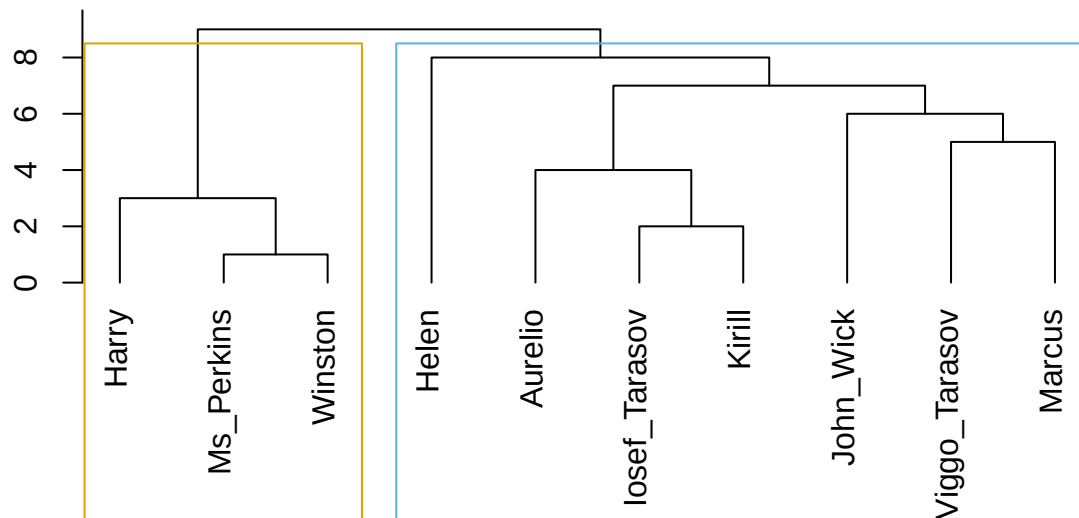
```
jw.comm.wt <- jw.g %>% cluster_walktrap()
modularity(jw.comm.wt)
```

```
## [1] 0.125
```

```
plot(jw.comm.wt, jw.g, layout=jw.layout)
```



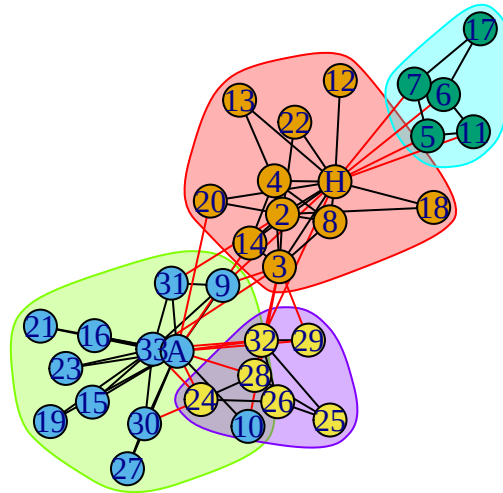
```
jw.comm.wt %>% plot_dendrogram()
```



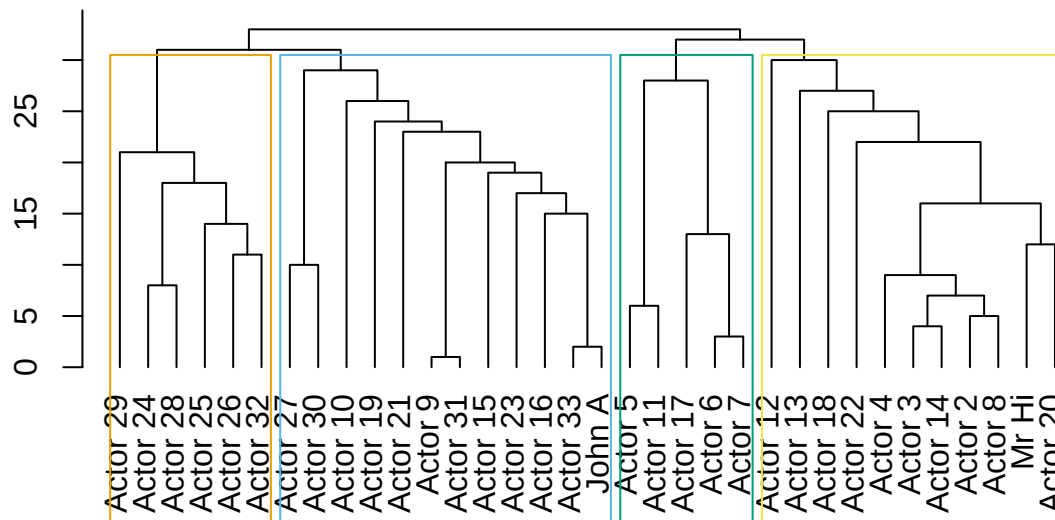
```
karate.comm.wt_2 <- karate.g %>% cluster_walktrap(steps = 2)
modularity(karate.comm.wt_2)
```

```
## [1] 0.4449036
```

```
plot(karate.comm.wt_2, karate.g, layout=karate.layout)
```



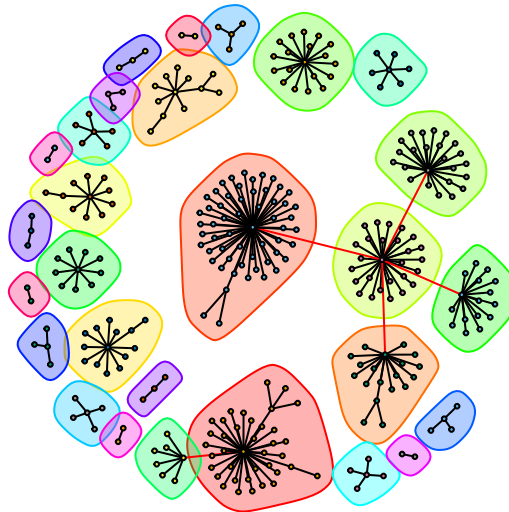
```
karate.comm.wt_2 %>% plot_dendrogram()
```



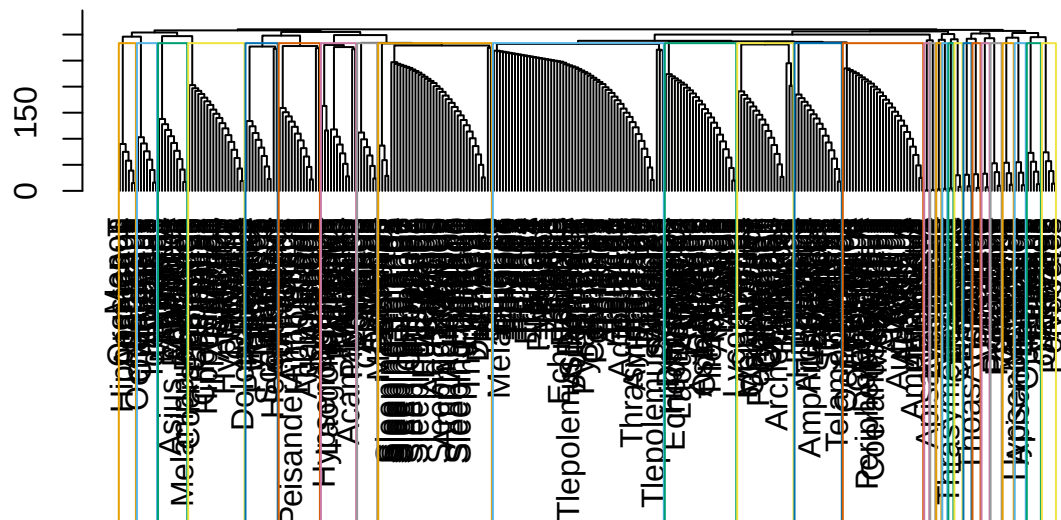
```
iliad.comm.wt_2 <- iliad.g %>% cluster_walktrap()
modularity(iliad.comm.wt_2)
```

```
## [1] 0.8922707
```

```
plot(iliad.comm.wt_2,
      iliad.g,
      layout=iliad.layout,
      vertex.label=NA,
      vertex.size=2)
```



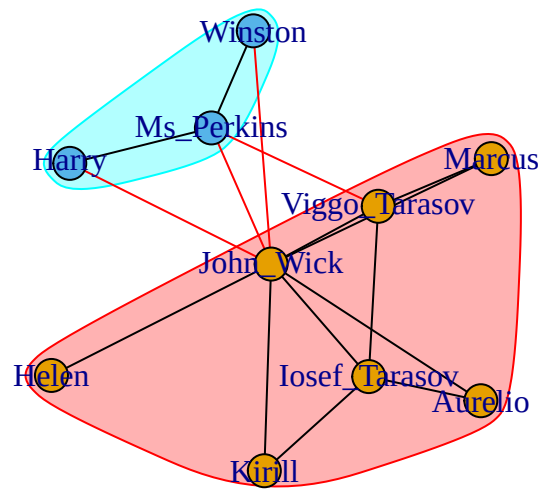
```
iliad.comm.wt_2 %>% plot_dendrogram()
```



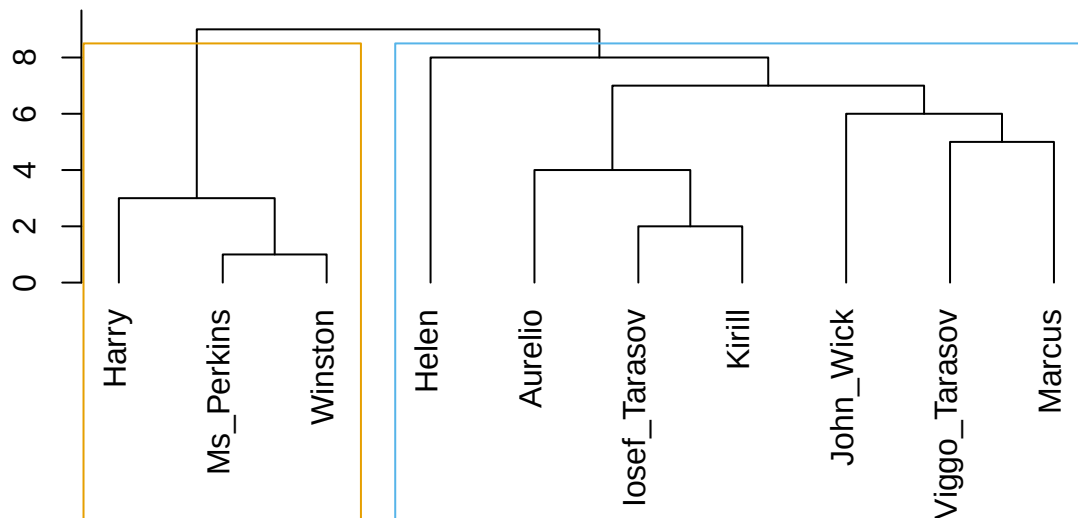
```
jw.comm.wt_2 <- jw.g %>% cluster_walktrap()
modularity(jw.comm.wt_2)
```

```
## [1] 0.125
```

```
plot(jw.comm.wt_2, jw.g, layout=jw.layout)
```



```
jw.comm.wt_2 %>% plot_dendrogram()
```



## Edge Betweenness (Girvan-Newman)

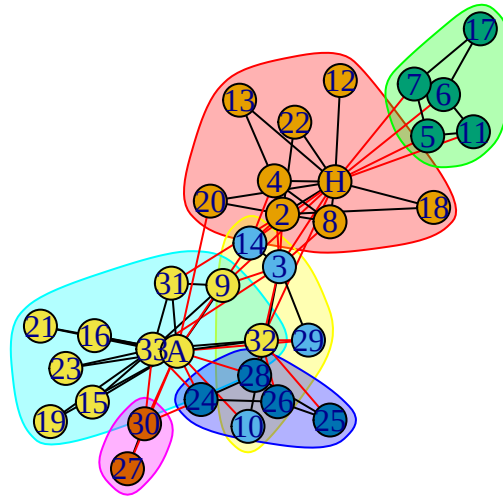
```
karate.comm.eb <- karate.g %>% cluster_edge_betweenness()
```

```
## Warning in cluster_edge_betweenness(.): At
## vendor/cigraph/src/community/edge_betweenness.c:497 : Membership vector will be
## selected based on the highest modularity score.
```

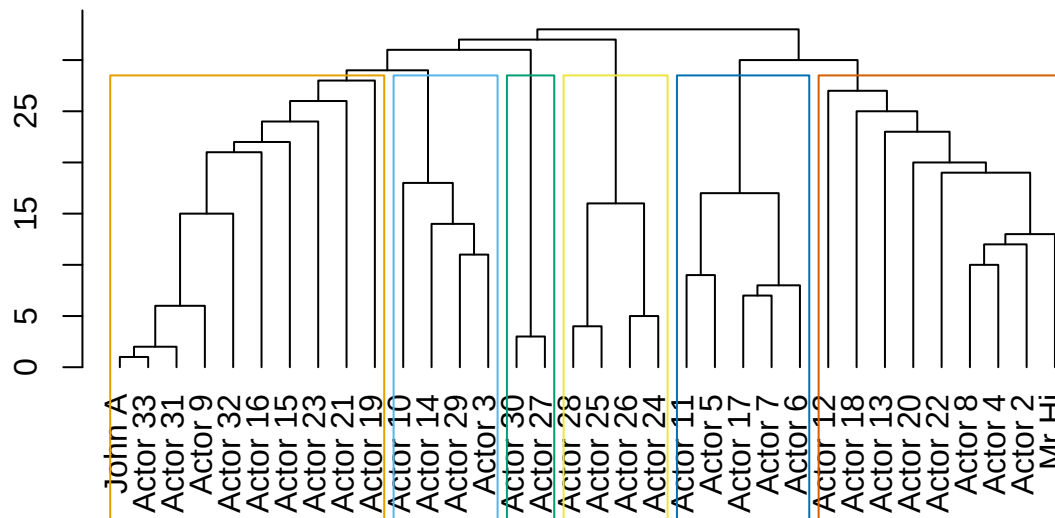
```
modularity(karate.comm.eb)
```

```
## [1] 0.345299
```

```
plot(karate.comm.eb, karate.g, layout=karate.layout)
```



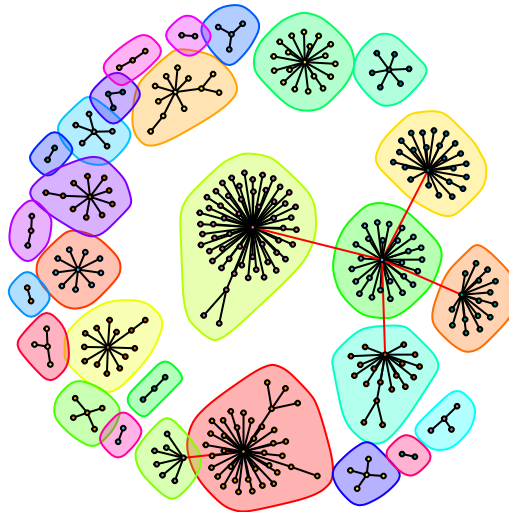
```
karate.comm.eb %>% plot_dendrogram()
```



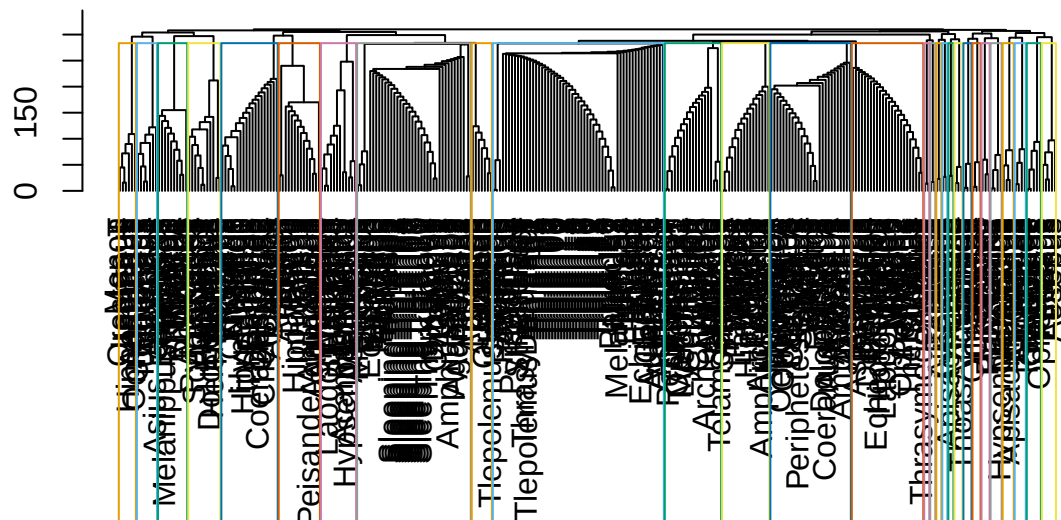
```
iliad.comm.eb <- iliad.g %>% cluster_edge_betweenness()
modularity(iliad.comm.eb)
```

```
## [1] 0.8922707
```

```
plot(iliad.comm.eb,
      iliad.g,
      layout=iliad.layout,
      vertex.label=NA,
      vertex.size=2)
```



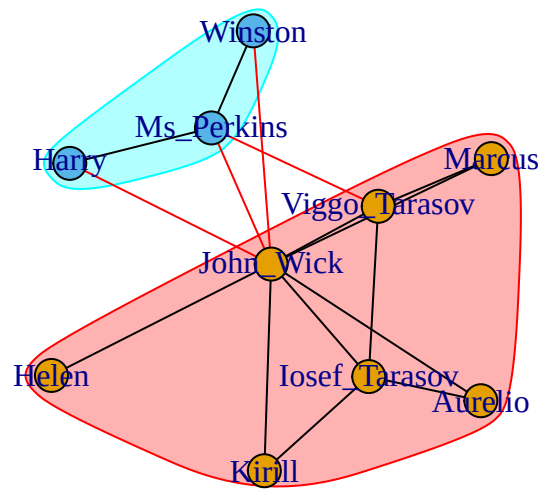
```
iliad.comm.eb %>% plot_dendrogram()
```



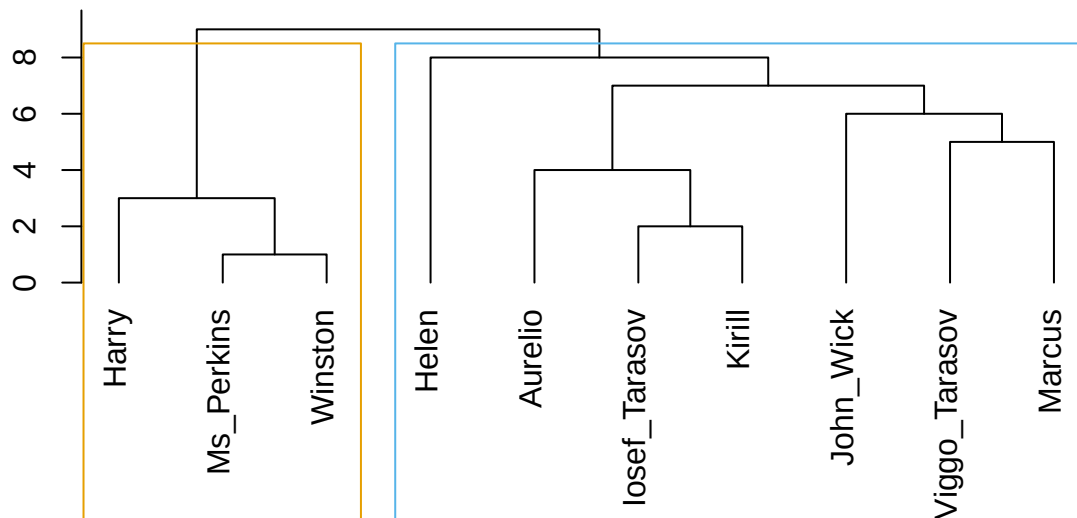
```
jw.comm.eb <- jw.g %>% cluster_edge_betweenness()  
modularity(jw.comm.eb)
```

```
## [1] 0.1074219
```

```
plot(jw.comm.wt_2, jw.g, layout=jw.layout)
```



```
jw.comm.wt_2 %>% plot_dendrogram()
```



```
# I was mistaken in class when i tried regular (node) betweenness.
# it is actually based on *edge* betweenness
jw.between <- edge_betweenness(jw.g)
jw.between
```

```
## [1] 5 7 4 7 7 5 7 7 9 2 3 2 2 3 2 2
```

```
E(jw.g)
```

```
## + 16/16 edges from f1cdf5c (vertex names):
## [1] John_Wick --Iosef_Tarasov John_Wick --Aurelio
## [3] John_Wick --Viggo_Tarasov John_Wick --Marcus
## [5] John_Wick --Kirill John_Wick --Ms_Perkins
## [7] John_Wick --Harry John_Wick --Winston
## [9] John_Wick --Helen Iosef_Tarasov--Aurelio
## [11] Iosef_Tarasov--Viggo_Tarasov Iosef_Tarasov--Kirill
## [13] Viggo_Tarasov--Marcus Viggo_Tarasov--Ms_Perkins
## [15] Ms_Perkins --Harry Ms_Perkins --Winston
```

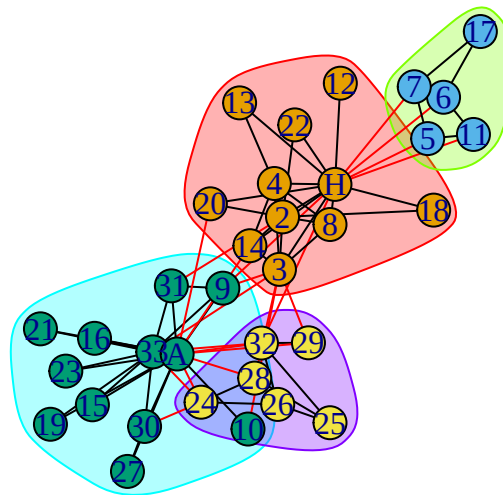
## Louvain

Agglomerative. Start with no communities. Add each node to community until local maximum for modularity.

```
karate.comm.louvain <- karate.g %>% cluster_louvain()
modularity(karate.comm.louvain)
```

```
## [1] 0.4449036
```

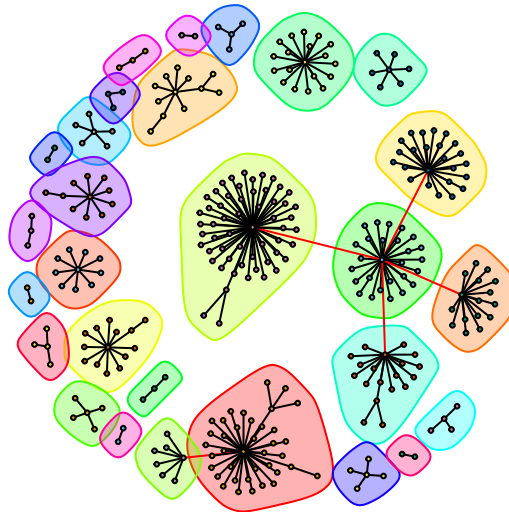
```
plot(karate.comm.louvain, karate.g, layout=karate.layout)
```



```
iliad.comm.louvain <- iliad.g %>% cluster_louvain()
modularity(iliad.comm.louvain)
```

```
## [1] 0.8922707
```

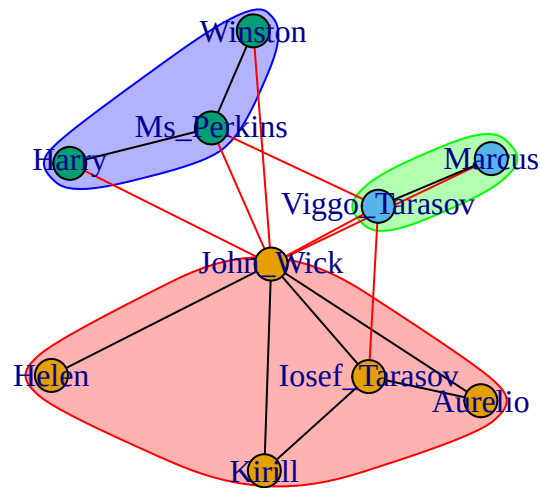
```
plot(iliad.comm.louvain,
     iliad.g,
     layout=iliad.layout,
     vertex.label=NA,
     vertex.size=2)
```



```
jw.comm.louvain <- jw.g %>% cluster_louvain()  
modularity(jw.comm.louvain)
```

```
## [1] 0.1484375
```

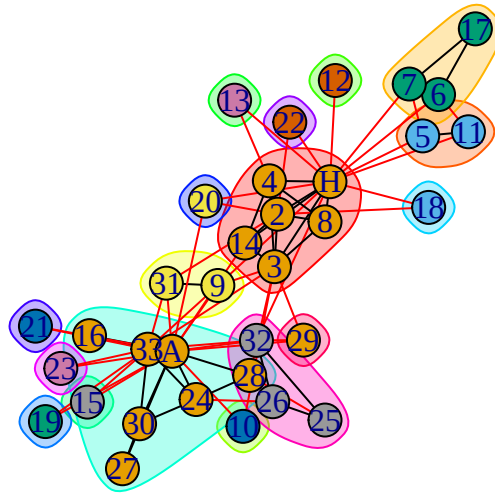
```
plot(jw.comm.louvain, jw.g, layout=jw.layout)
```



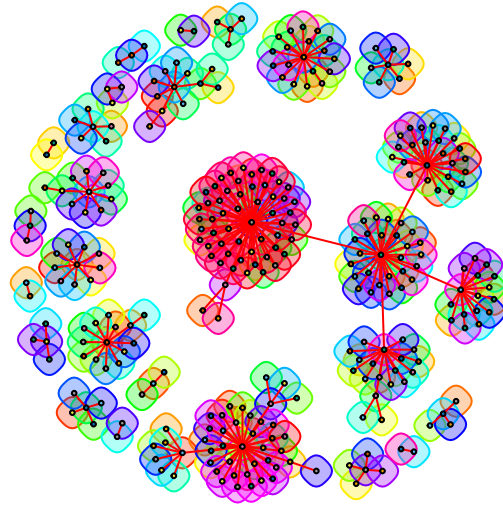
## Leiden

(Improved Louvain)

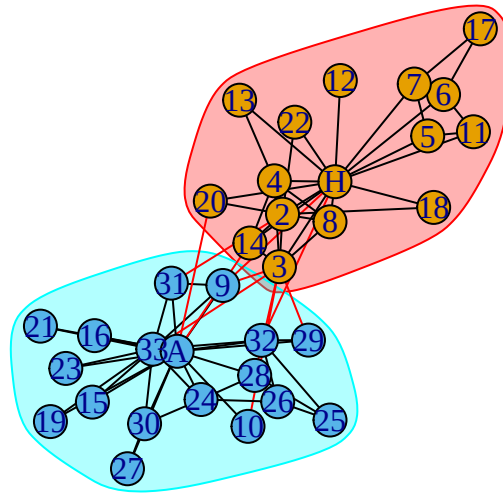
```
karate.comm.leiden <- karate.g %>% cluster_leiden()
# no modularity as defaults to Constant Potts Model
plot(karate.comm.leiden, karate.g, layout=karate.layout)
```



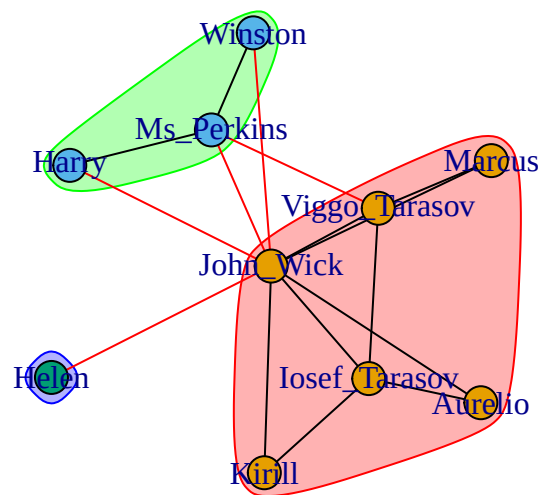
```
iliad.comm.leiden <- iliad.g %>% cluster_leiden()
#modularity(iliad.comm.leiden)
plot(iliad.comm.leiden,
      iliad.g,
      layout=iliad.layout,
      vertex.label=NA,
      vertex.size=2)
```



```
karate.comm.leiden_01 <- karate.g %>% cluster_leiden(resolution=0.1)
plot(karate.comm.leiden_01, karate.g, layout=karate.layout)
```



```
jw.comm.leiden_03 <- jw.g %>% cluster_leiden(resolution=0.3)
plot(jw.comm.leiden_03, jw.g, layout=jw.layout)
```



```
iliad.comm.leiden_001 <- iliad.g %>% cluster_leiden(resolution=0.01)
plot(iliad.comm.leiden_001,
     iliad.g,
     layout=iliad.layout,
     vertex.label=NA,
     vertex.size=2)
```

